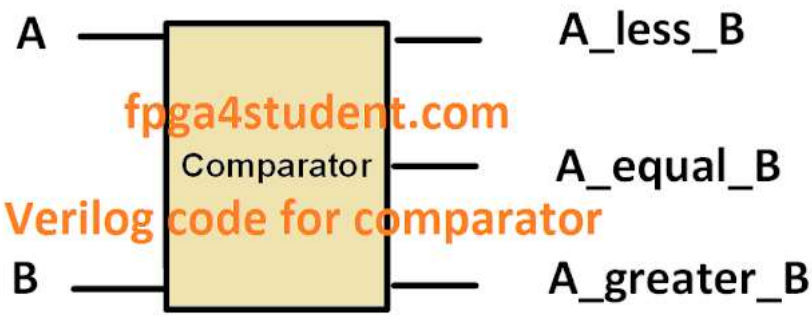
Verilog code for a comparator

In this project, a simple 2-bit comparator is designed and implemented in Verilog HDL. Truth table, K-Map and minimized equations for the comparator are presented. The Verilog code of the comparator is simulated by ModelSim and the simulation waveform is presented.



The specification of the 2-bit comparator is as follows:

- **Input: 2-bit A and B for comparison**
- **Output:**
  - **A_greater_B: high if A > B else low**
  - **A_equal_B: high if A = B else low**
  - **A_less_B: high if A<B else low**

The truth table for the comparator:

A1	A0	B1	B0	A_greater_B	A_equal_B	A_less_B
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

K-Map tables and the corresponding equations for the comparator:

A_greater_B	A1A0 00	01	11	10
B1B0 00	0	1	1	1
01	0	0	1	1
11	0	0	0	0
10	0	0	1	0

A_greater_B = B0 B1 A0 + B1 A1 + A1 A0 B0
= A0 B0 (B1 + A1) + + B1 A1

A_equal_B	A1A0 00	01	11	10
B1B0 00	1	0	0	0
01	0	1	0	0
11	0	0	1	0
10	0	0	0	1

A_equal_B = (B1 A1+A1 B1) (B0 A0+A0 B0)

A_less_B	A1A0 00	01	11	10
B1B0 00	0	0	0	0
01	1	0	0	0
11	1	1	0	1
10	1	1	0	0

A_less_B = A0 A1 B0 + A1 B1 + B1 B0 A0

After obtaining the minimized equations of the outputs of the comparator, it is easy to write Verilog code for the comparator.

Below is the Verilog code for the comparator:

```
// FPGA projects using Verilog/ VHDL
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// Verilog code for 2-bit comparator
module comparator(input [1:0] A,B, output A_less_B, A_equal_B, A_greater_B);
wire tmp1,tmp2,tmp3,tmp4,tmp5, tmp6, tmp7, tmp8;
// A = B output
xnor u1(tmp1,A[1],B[1]);
xnor u2(tmp2,A[0],B[0]);
and u3(A_equal_B,tmp1,tmp2);
// A less than B output
assign tmp3 = (~A[0])& (~A[1])& B[0];
assign tmp4 = (~A[1])& B[1];
assign tmp5 = (~A[0])& B[1]& B[0];
assign A_less_B = tmp3 | tmp4 | tmp5;
// A greater than B output
assign tmp6 = (~B[0])& (~B[1])& A[0];
assign tmp7 = (~B[1])& A[1];
assign tmp8 = (~B[0])& A[1]& A[0];
assign A_greater_B = tmp6 | tmp7 | tmp8;
endmodule
`timescale     10 ps/ 10 ps
// FPGA projects using Verilog/ VHDL
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// Verilog testbench code for 2-bit comparator
module tb_comparator;
reg [1:0] A, B;
wire A_less_B, A_equal_B, A_greater_B;
integer i;
// device under test
comparator dut(A,B,A_less_B, A_equal_B, A_greater_B);
initial begin
     for (i=0;i<4;i=i+1)
     begin
```
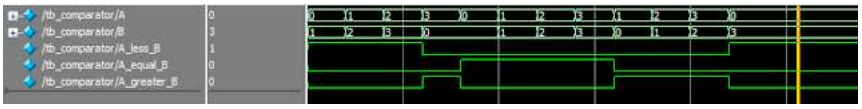

```
            A = i;
            B = i + 1;
            #20;
        end
        for (i=0;i<4;i=i+1)
        begin
            A = i;
            B = i;
            #20;
        end
        for (i=0;i<4;i=i+1)
        begin
            A = i+1;
            B = i;
            #20;
        end
    end
    endmodule
```

Finally, simulate the Verilog code and testbench in ModelSim and verify the operation of the comparator through simulation waveform.



By observing the waveform, Verilog code for the comparator are working correctly.

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